ESL

مختبر الإمارات للسلامة
EMIRATES SAFETY LABORATORY

TEST REPORT

REPORT NUMBER.
0206-25-CR-01

Reaction to Fire Classification Report of 4mm thick “Alubond USA FR A2” Aluminium Composite Panel

According to: EN 13501-1:2018

Date of issue: 02 October 2025

ilac-MRA IAS ACCREDITED Testing Laboratory TL-1038 egolf

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1 Introduction

This classification report defines the classification assigned to 4mm thick “Alubond USA FR A2” Aluminium Composite Panel in accordance with the procedures given in EN 13501-1:2018.

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CLASSIFICATION OF REACTION TO FIRE
IN ACCORDANCE WITH EN 13501-1:2018

Sponsor:	Eurocon Building Industries FZE Plot 2E-03, 2E-08, Hamriyah Freezone, Sharjah, United Arab Emirates
Prepared by:	Emirates Safety Laboratory Warsan Third, Dubai, United Arab Emirates
Notified Body No:	Not applicable
Test performed at:	Emirates Safety Laboratory Warsan Third, Dubai, United Arab Emirates
Product name:	4mm thick “Alubond USA FR A2” Aluminium Composite Panel
Classification report no.	0206-25-CR-01
Issue number:	1
Date of issue:	02 October 2025

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2 Details of the Classified Product

2.1 General

The product is defined by the test sponsor as Aluminium Composite Panel.

2.2 Product Description

The product, 4mm thick “Alubond USA FR A2” Aluminium Composite Panel is described in support of the classification listed in Section 4.

Note:
1: The test sponsor does not want to disclose the information; thus, this was kept confidential by ESL.
[A]: The information is as declared by the client and ESL Certification; ESL Testing was unable to verify the information.
[B]: The information is as measured by ESL Testing.

General description	Aluminium Composite Panel (ACP)	
Product name	"Alubond USA FR A2" [A]	
Manufacturer	Eurocon Building Industries FZE Plot 2E-03, 2E-08, Hamriyah Freezone, Sharjah, United Arab Emirates	
Thickness	4mm[A]; 4.19mm[B]	
Area weight	8.2 ± 0.5 kg/m^{2}[A]; 8.8 kg/m^{2}[B] *(measured by the laboratory from a 250mm x 250mm specimen)*	
Topcoat	Type	Polyvinylidene fluoride (PVDF)[A]
	Manufacturer	Spectrum Industries (L.L.C)[A]
	Worst colour tested	Black – as chosen by ESL Certification from Table 2 of EN ISO 1716 report (0206-25-TR-01)
	Thickness	18-20microns[A]
	Density	1700kg/m^{3}[A]
Primer	Type	Polyvinylidene fluoride (PVDF)[A]
	Manufacturer	Spectrum Industries (L.L.C)[A]
	Thickness	6-7microns[A]
	Density	1170 kg/m^{3}[A]
Aluminium sheet	Description	Mill finish aluminium coil[A]
	Thickness	0.5 ± 0.05mm[A]
	Density	2720 kg/m^{3}[A]
Adhesive	Description	Polymer based adhesive film[A]
	Manufacturer	Alubond Middle East FZE
	Thickness	80microns[A]
	Density	938 ± 20 kg/m^{3}[A]
Core	Description	Mineral core[A]
	Manufacturer	Metalplast Industries FZE[A]
	Thickness	3.0 ± 0.1mm[A]; ; 3.10mm[B]
	Density	1850 ± 100 kg/m^{3}[A]; 1845 kg/m^{3}[B]
Adhesive	Description	Polymer based adhesive film[A]
	Manufacturer	Alubond Middle East FZE
	Thickness	80microns[A]
	Density	938 ± 20 kg/m^{3}[A]
Aluminium sheet	Description	Mill finish aluminium coil[A]
	Thickness	0.5 ± 0.05mm[A]
	Density	2720 kg/m^{3}[A]
Service coat	Type	Polyester (PE) Monocoat[A]
	Manufacturer	Spectrum Industries (L.L.C)[A]

	Thickness	8-10 microns[A]
	Density	1100 kg/m^3[A]
Dimensions of the test specimens as measured by the laboratory, see Section 4.3 *(Each specimen consisted of two wings mounted into an aperture in a specimen trolley forming a vertical 90° corner)*	Long wing (width x height)	
	Panel A	792.5 x 992.5 mm
	Panel B	196.5 x 992.5 mm
	Panel C	792.5 x 492.5 mm
	Panel D	196.5 x 492.5 mm
	Short wing	495 x 1500mm (width x height)
Backing board	Material and classification	Calcium Silicate board *A2-s1, d0 based on EN 13501-1:2018*
	Thickness, mm	12
	Density, kg/m^3	870

2.3 Reports

Name of laboratory	Name of sponsor	Report ref. no.	Test method and date
Emirates Safety Laboratory	Eurocon Building Industries FZE	0206-25-TR-01 02 October 2025	EN ISO 1716:2018
		0206-25-TR-02 02 October 2025	EN 13823:2020+A1:2022

2.4 Results

Test method and test number	Parameter	No. of tests	*TEST RESULTS*	
			Continuous parameter - mean	Compliance with parameters
EN ISO-1716:2018 (0206-25-TR-01)	*External non-substantial layer (Topcoat + Primer)* Q_{pcs} MJ/m^2 ≤ 4.0 MJ/m^2	3	0.7	Compliant
	External non-substantial layer (Back coat) Q_{pcs} MJ/m^2 ≤ 4.0 MJ/m^2		0.2	
	Substantial Layer (Core)		1.7	

	Q_{pcs} MJ/kg ≤ 3.0 MJ/kg			
	Substantial Layer (Aluminium) Q_{pcs} MJ/kg ≤ 3.0 MJ/kg		0	
	Internal non-substantial layer (Adhesive) Q_{pcs} MJ/m^2 ≤ 4.0 MJ/m^2		3.5	
	Product as a whole Q_{pcs}MJ/kg ≤ 3.0 MJ/kg		2.0	
EN 13823:2020+A1:2022 (0206-25-TR-02)	**For Class "A2"**			
	THR $_{600s}$ ≤ 7.5 MJ	3	0.7	Compliant
	FIGRA$_{0.2MJ}$ ≤ 120 W/s	3	1	Compliant
	Lateral flame spread < Edge of specimen	3	< Edge of specimen	Compliant
	For subclass "s1"			
	TSP$_{600s}$ ≤ 50 m$^{2(B)}$	3	11	Compliant
	SMOGRA ≤ 30m^2/s$^{2(B)}$	3	0	Compliant
	For subclass "d0"			
	No flaming droplets/ particles within 600s	3	Not observed	Compliant

3 Classification and Field of Application

3.1 Reference of classification

This classification has been carried out in accordance with EN 13501-1:2018.

3.2 Classification

The product, 4mm thick "Alubond USA FR A2" Aluminium Composite Panel, in relation to its reaction to fire behaviour is classified as:

A2

The additional classification in relation to smoke production is:

s1

The additional classification in relation to flaming droplets/particles is:

d0

The format of the reaction to fire classification for construction products excluding floorings and linear pipe thermal insulation products is:

Fire behaviour		Smoke production			Flaming droplets	
A2	-	s	1	,	d	0

Reaction to fire classification: A2-s1, d0

3.3 Field of application

The classification is valid for the following product parameters:

Colour of topcoat	Any variation in the tested type
Type of topcoat and primer	No variation allowed
Thickness and area density of the components	No variation allowed
Thickness and area density of the product	No variation allowed
Product composition and layout	No variation allowed
Joints	For assembly without joints, and with joints up to the tested configuration
Air gap	Ventilated

This classification is valid for the following end-use applications:
Details of buildup/Construction - No variation in the build-up allowed.

3.4 Limitations

This report does not represent type approval or certification of the product.

4 Signatories

Prepared by	*Reviewed by*	*Authorised by*
Rachel Marie Novelo Reaction to Fire Principal Engineer	**Sebastian Ukleja** Testing Manager	**Sebastian Ukleja** Testing Manager
Signature	Signature	Signature
Rnovela	Uklega	Uklega

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